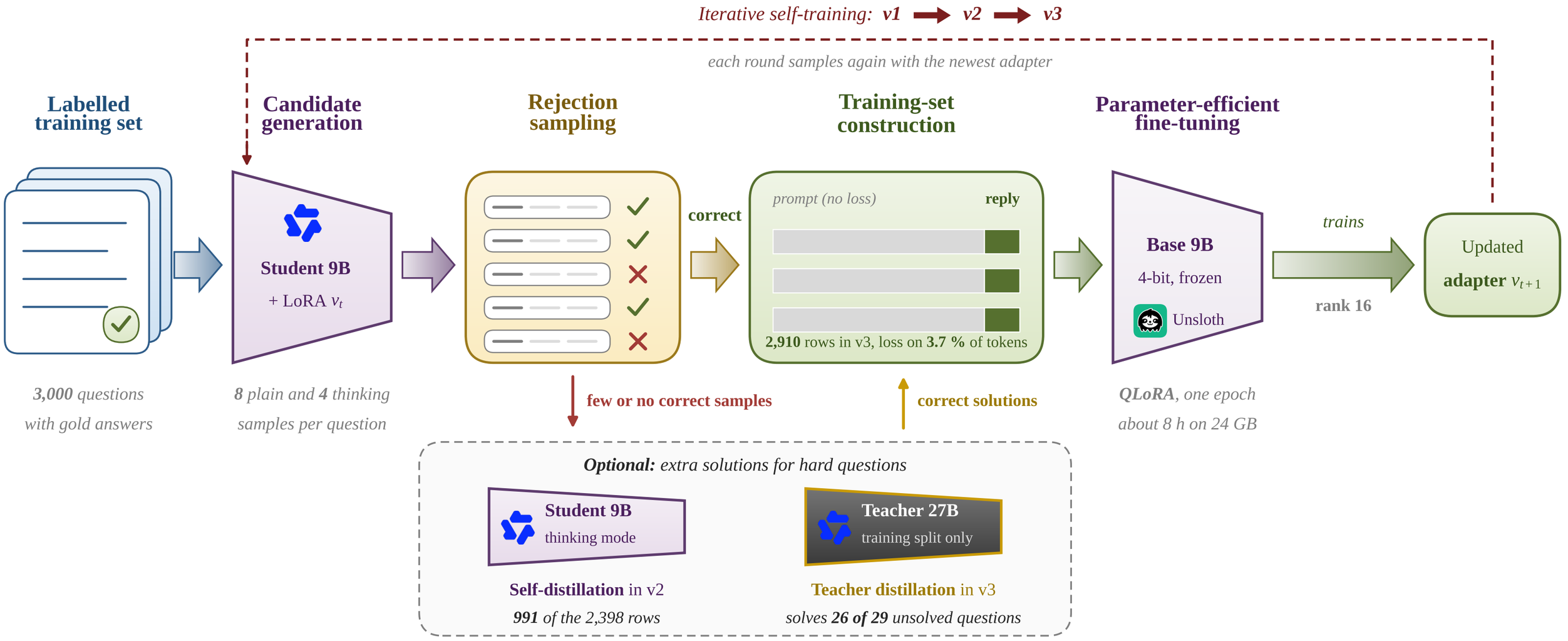

Iterative self-training: v1 → v2 → v3
each round samples again with the newest adapter
Labelled training set
Candidate generation
Rejection sampling
Training-set construction
Parameter-efficient fine-tuning
Student 9B
+ LoRA v_t
correct
prompt (no loss)
reply
2,910 rows in v3, loss on 3.7 % of tokens
Base 9B
4-bit, frozen
Unsloth
trains
rank 16
Updated adapter v_{t+1}
3,000 questions with gold answers
8 plain and 4 thinking samples per question
few or no correct samples
correct solutions
QLoRA, one epoch about 8 h on 24 GB
Optional: extra solutions for hard questions
Student 9B
thinking mode
Self-distillation in v2
991 of the 2,398 rows
Teacher 27B
training split only
Teacher distillation in v3
solves 26 of 29 unsolved questions